

Lecture #2

Basic Economics Models

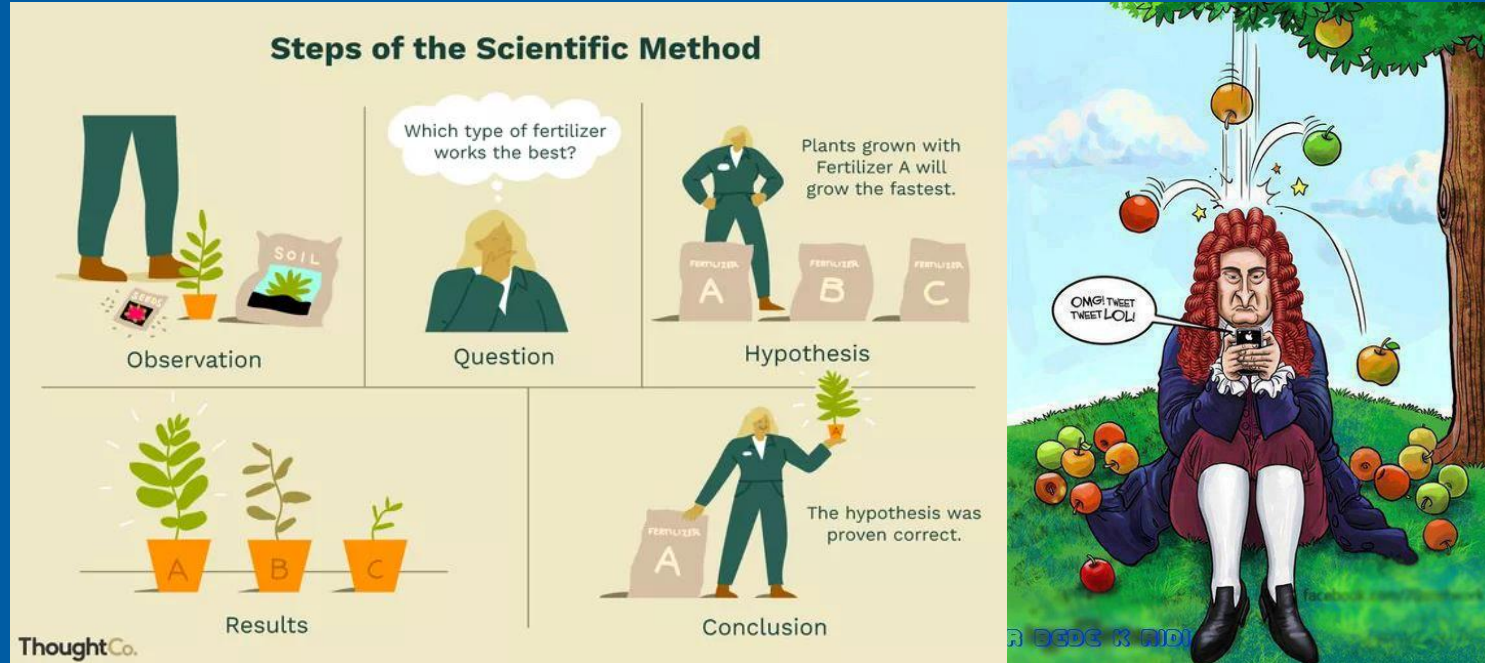
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The Scientific Method



REMEMBER: Economics is a *Social* Science

The Scientific Method

The scientific method involves identifying a problem, gathering data, forming a hypothesis, testing the hypothesis, and analyzing the results.

1. Observation
 - An apple hit me on the head 🍏🤔
 - When Molde FK plays at home, it is more likely to win ⚽
2. Question
 - What led the apple to fall?
 - Why Molde FK is more likely to win at home?
3. Hypothesis
 - A force drives the apple (i.e., gravity)
 - A force drives Molde FK (i.e., supporters)
4. Experiment
 - Understand which variables come into play and their role
5. Results
 - Data visualization
6. Conclusion



Role of Assumptions

Economists make assumptions in order to simplify the complex world and make it easier to understand.

Economists use different assumptions to answer different questions.

- For studying short-run effects we may assume that prices do not change much.
 - Ticket prices fixed
- For studying long-run effects we may assume that prices are completely flexible.
 - Ticket prices renegotiated between seasons



Role of Assumptions



A physicist, a chemist and an economist are stranded on an island, with nothing to eat. A can of soup washes ashore.

The physicist says, "Let's smash the can open with a rock."

The chemist says, "Let's build a fire and heat the can first."

The economist says, "Let's assume that we have a can-opener..."

--Paul Samuelson



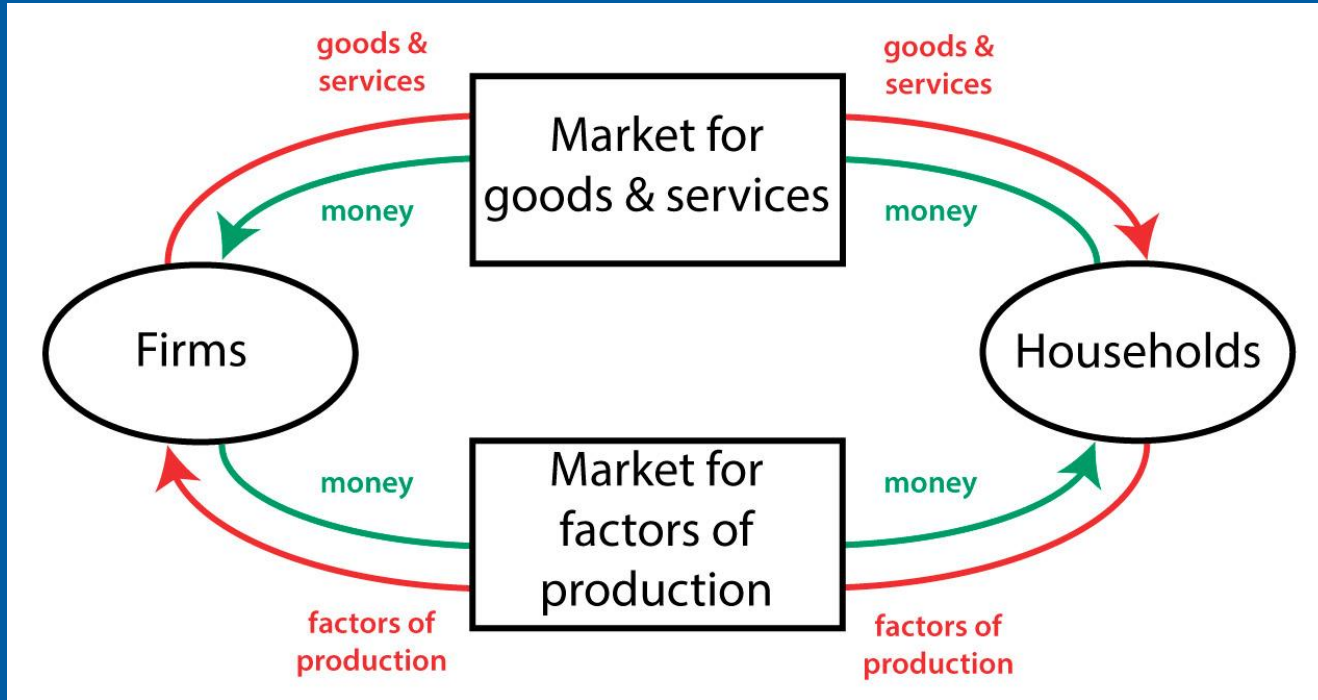
Economic Models

Two of the most basic models in economics are :

1. The Circular Flow Diagram
2. The Production Possibilities Frontier



The Circular-Flow Diagram



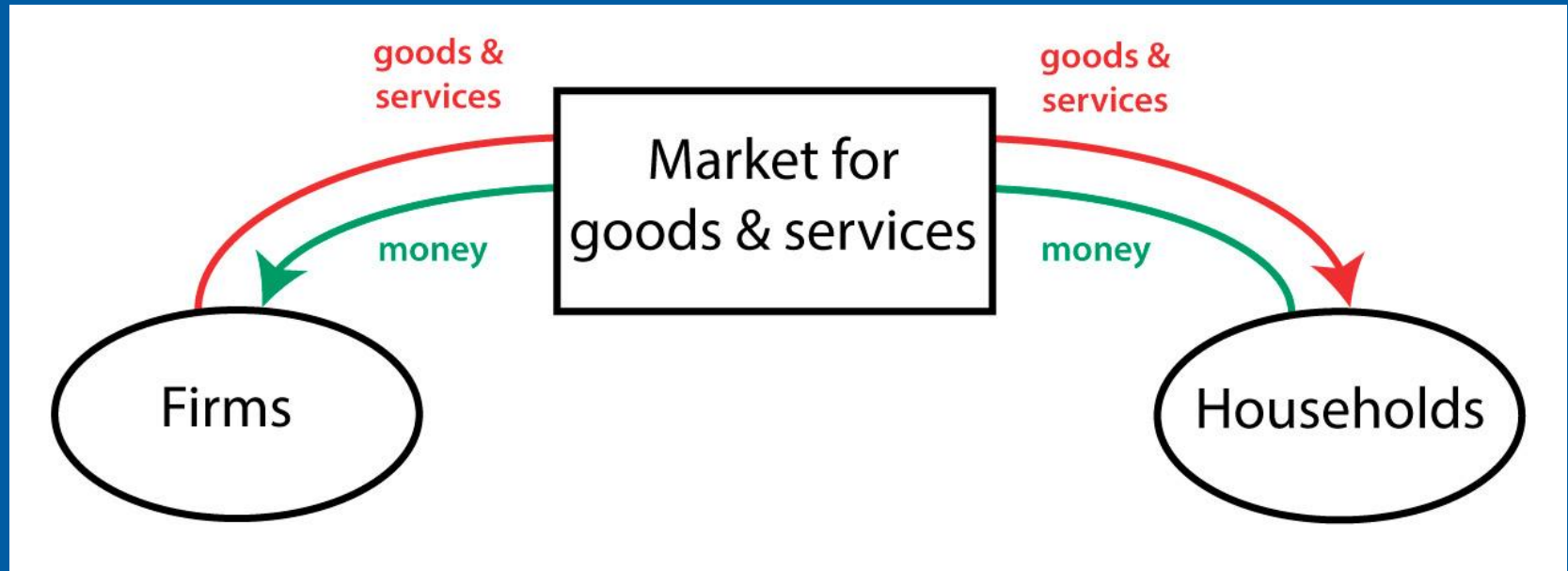
The Circular-Flow Diagram

The circular-flow diagram (or circular-flow model) depicts the flows of goods and money between two distinct parts of the economy:

1. Goods and Services Market
2. Factors of Production Market



Goods and Services Markets

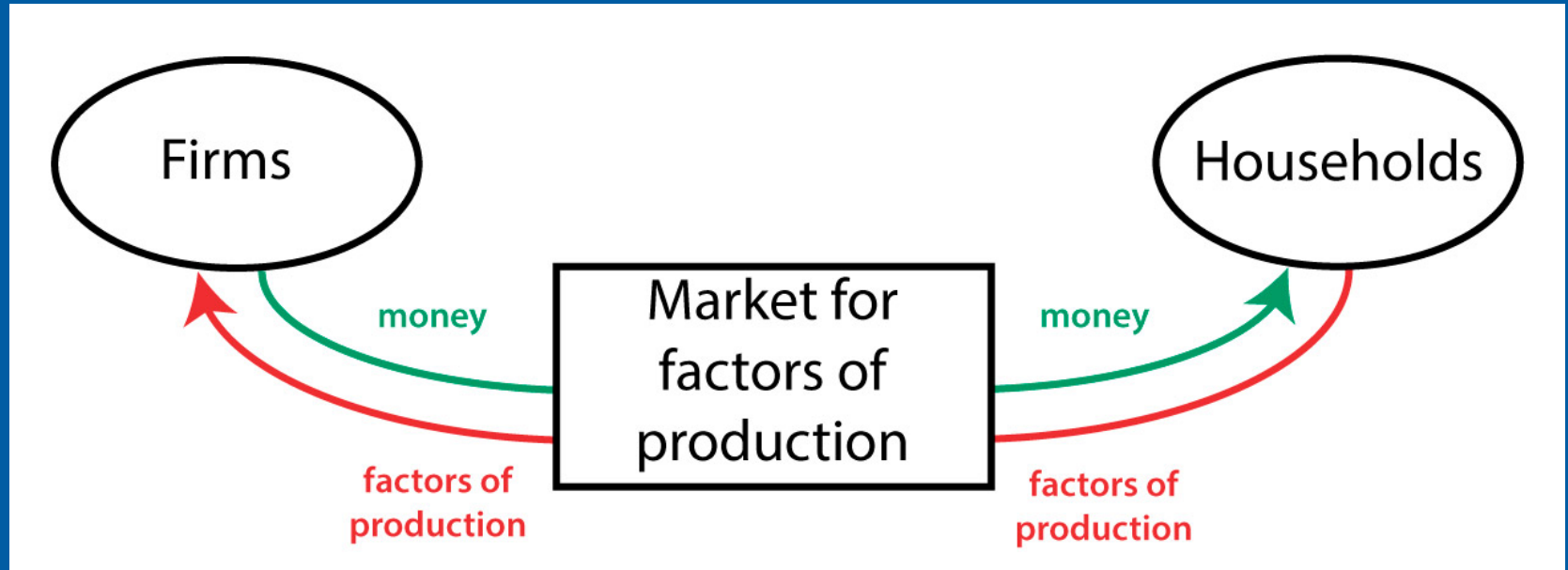


Goods and Services Market

- Households buy finished goods and services from firms in exchange for money.
- It is the place where firms sell the goods and services they have produced, receiving a revenue paid by households
- If the goods and services market was the only existing market, what would happen?
 - Firms would end up having all the money, while households would have all the finished products. In this case, economic activity would stop. But this is just half-story!
- Ex: Coop, Rema1000, gas stations, Burger King, Ikea, dentist....



Factors Market

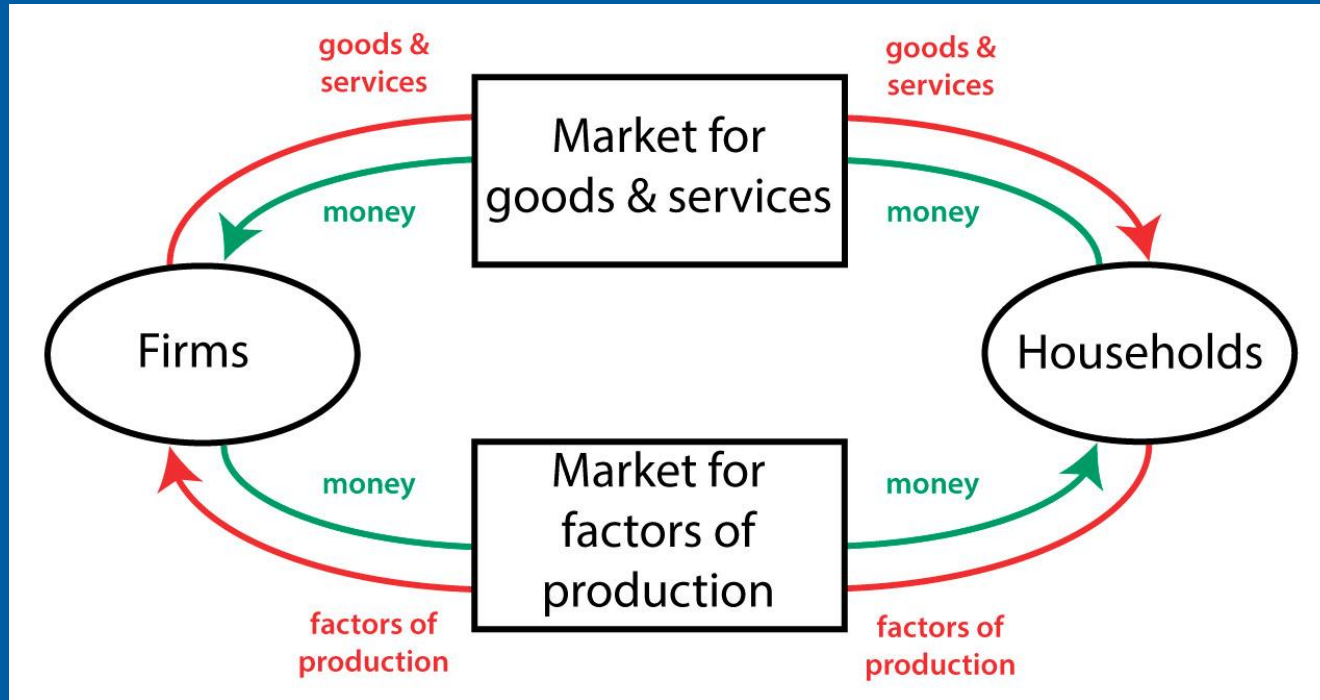


Factors Market

- «Factors of production» refers to anything that is used by a firm in order to make a final product
- Some examples:
 - Labor = work done by households, athletes, coaches, staff
 - Capital = machines used to make products, funds used to buy machines, stadiums/training equipment
 - Land
- Households offer their labour, capital, and land in exchange of money (in the form of income)



Circular-Flow Diagram (Full Picture)

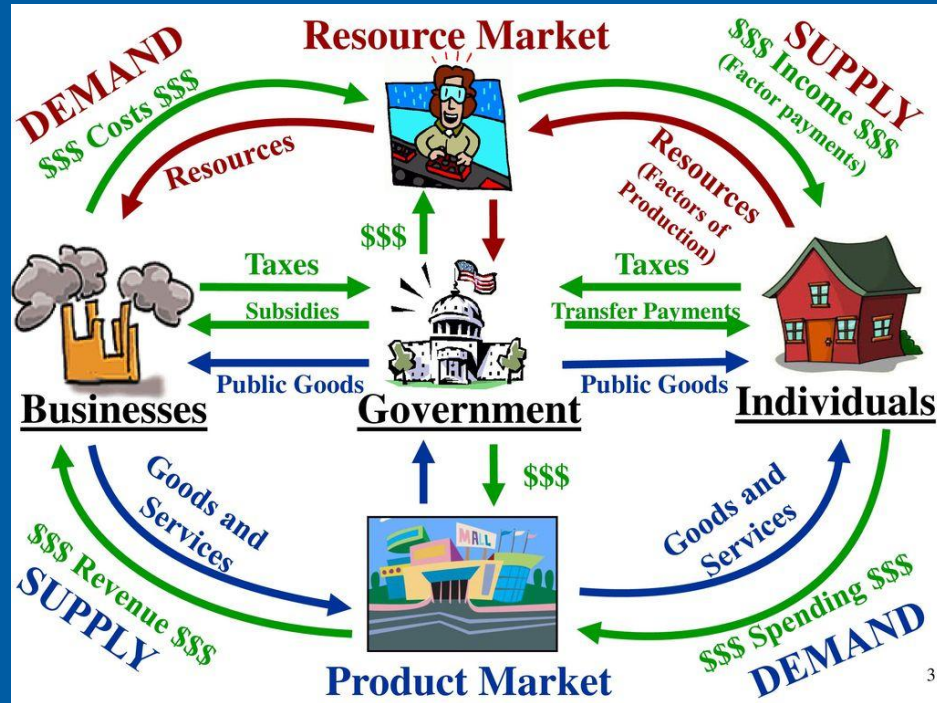


Circular-Flow Diagram (Full Picture)

- When we combine both diagrams, we get the full picture.
- A close loop for the flow is formed. As a result, continued economic activity is sustainable in the long run, since neither firms nor households are going to end up with all the money.
- In this basic model, firms use factors of production to create finished products and households consume finished products in order to maintain their ability to provide factors of production.
- It is an oversimplified model representing a purely capitalistic economy (only production, distribution, and consumption are considered).



Circular-Flow Diagram (Advanced)



Circular-Flow Diagram (Advanced)

- We could extend the model to include government intervention.
 - Income tax = link between households and factors market
 - Tax on producer = link between firms and goods and services market



Why Circular-Flow Diagram is important?

- It gives us the first grasp of supply and demand model
 - Goods and services market
 - Demand: households
 - Supply: firms
 - Factors market
 - Demand: firms
 - Supply: households



Production Possibilities Model

- Unlimited wants, need, desires, and dreams. But limited resources (such as time and money) 🕒 🌱 → It becomes critical to allocate our (scarce) resources among alternative choices
- The Production Possibilities Frontier is an analytical tool which is used to illustrate and explain this problem of choice.
- The Production Possibilities Frontier (PPF) is a graph that shows all possible combinations of two goods when an economy is producing at full potential.
 - Simplified model, since it assumes only two goods are produced.
 - It only shows what can be produced – not what is consumed.

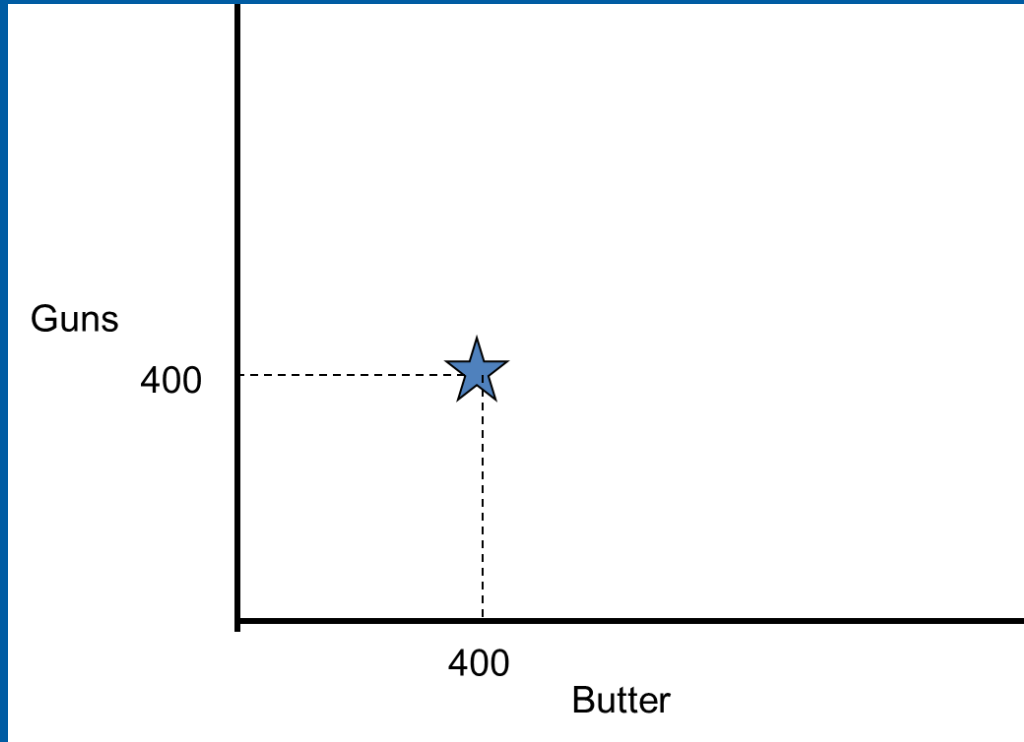


Assumptions

- Only two goods are produced in different proportions in the economy
- The same resources can be used to produce either or both two goods and can be shifted freely between them
- The supplies of factors are fixed. But they can be re-allocated to produce the two goods within limits.
- The production techniques are given and constant.
- The economy's resources are fully employed and technically efficient.
- The time period is short



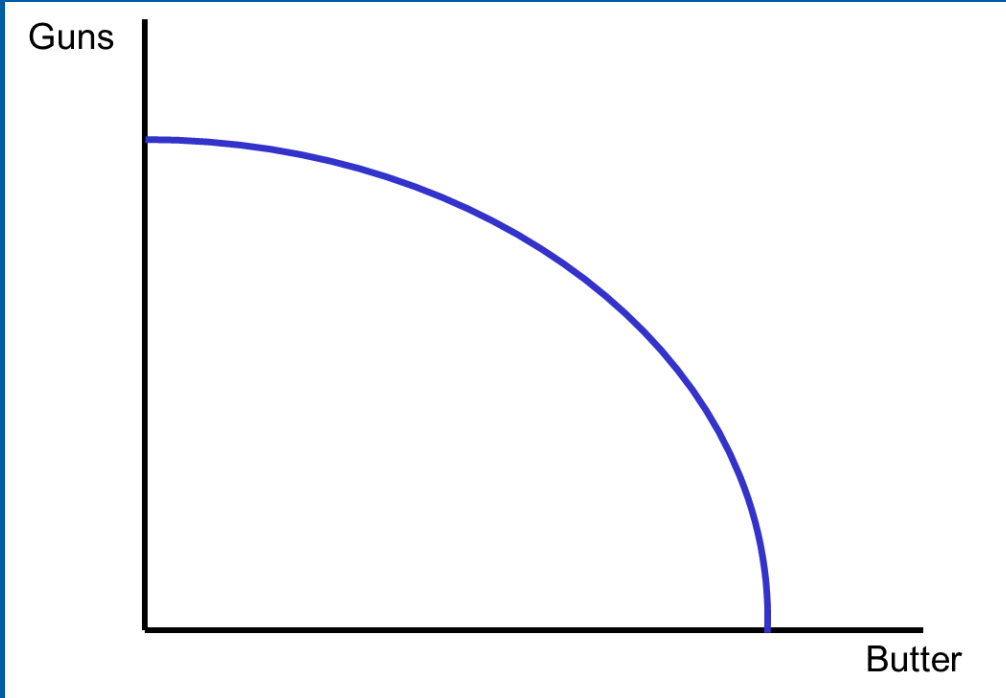
Production Possibilities Model



A point on the graph represents how much of each good is being produced.



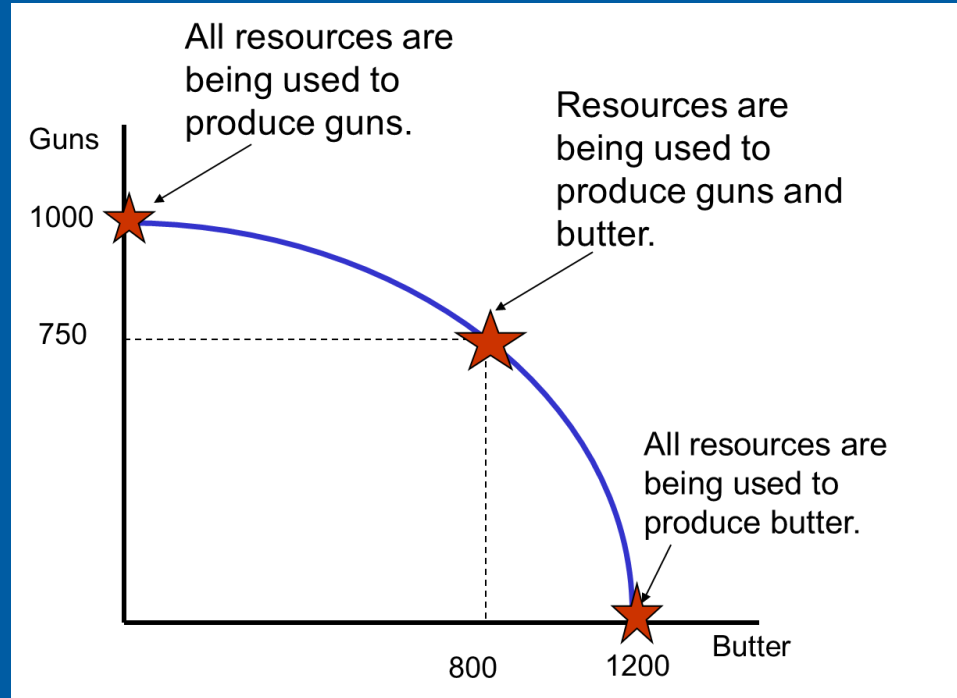
Production Possibilities Model



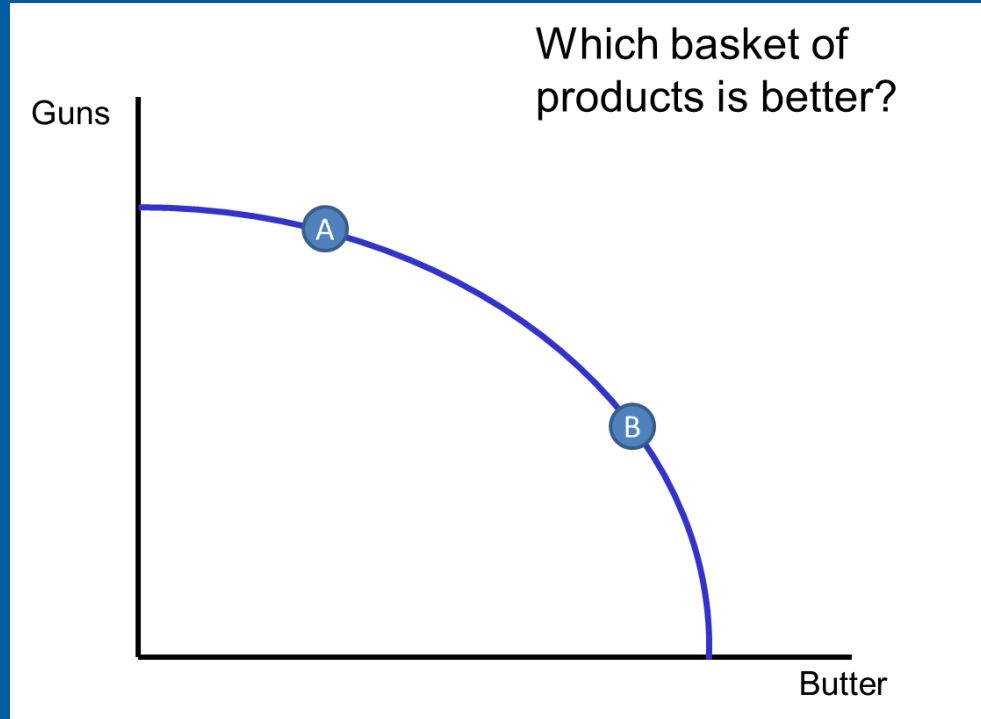
The frontier shows all **possible** combinations when all resources are fully employed.



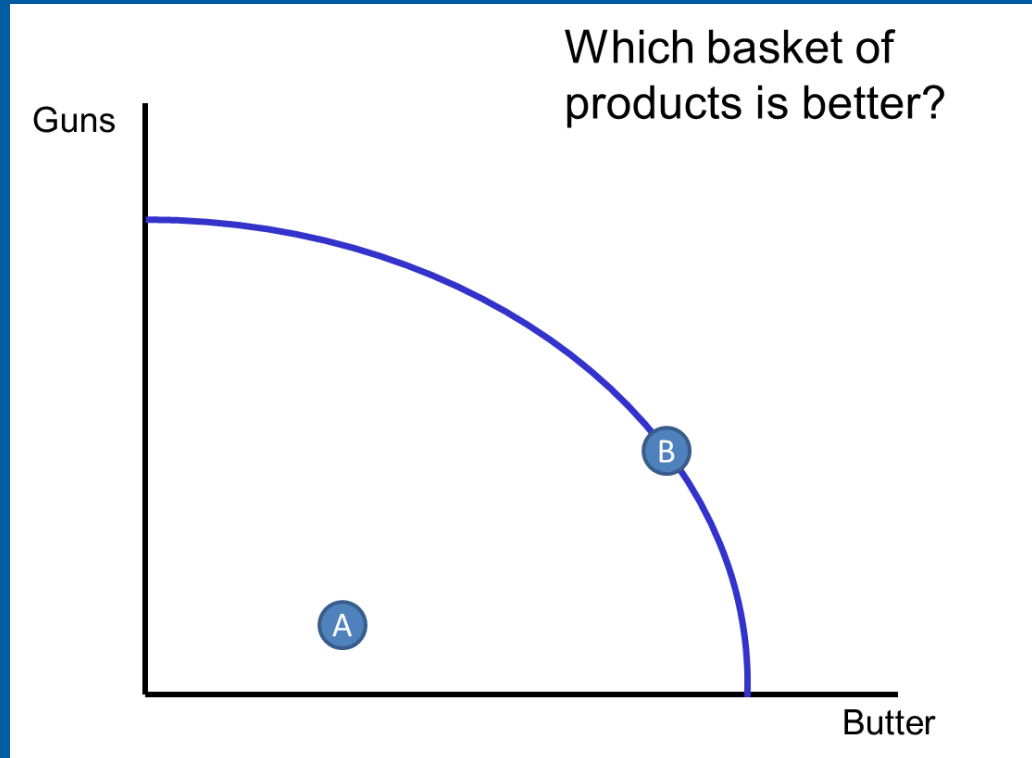
Production Possibilities Model



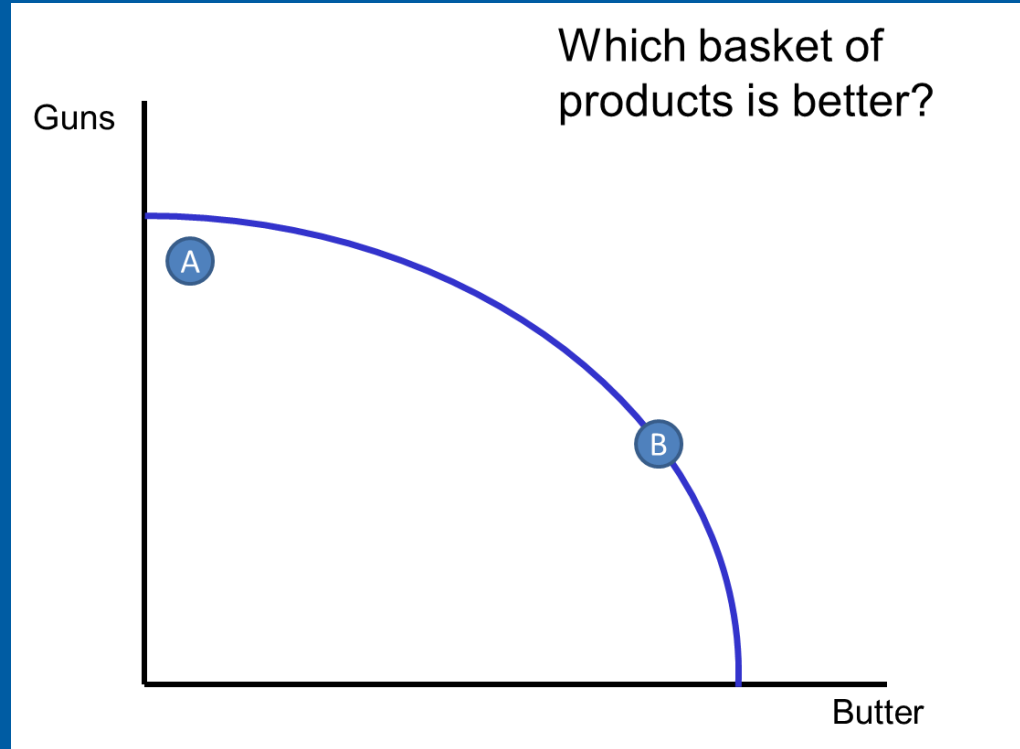
Production Possibilities Model



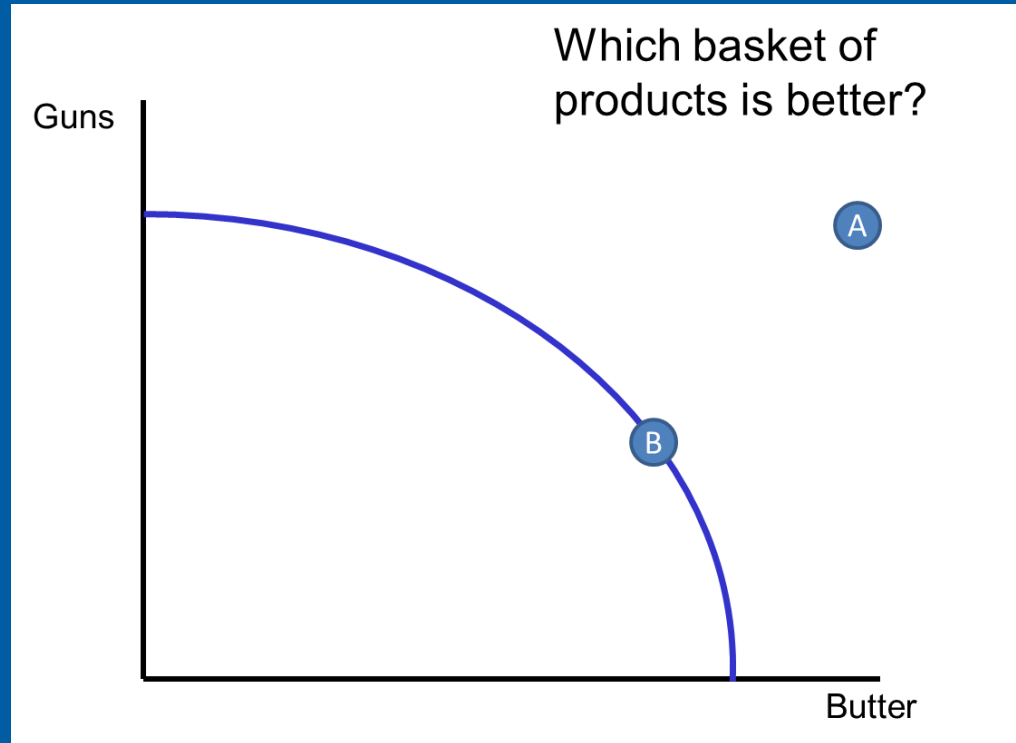
Production Possibilities Model



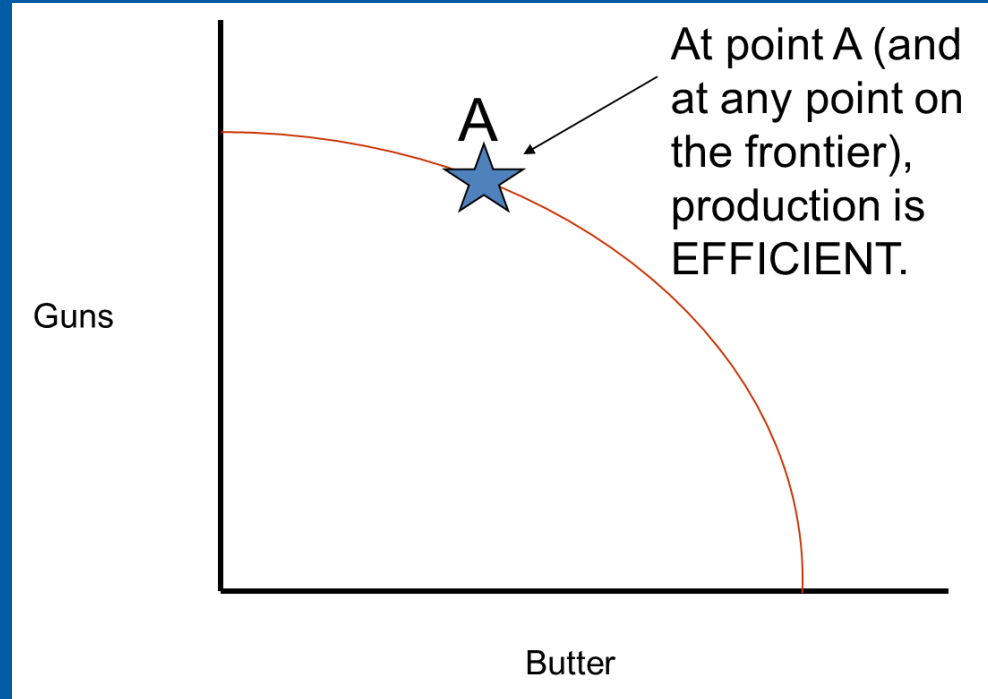
Production Possibilities Model



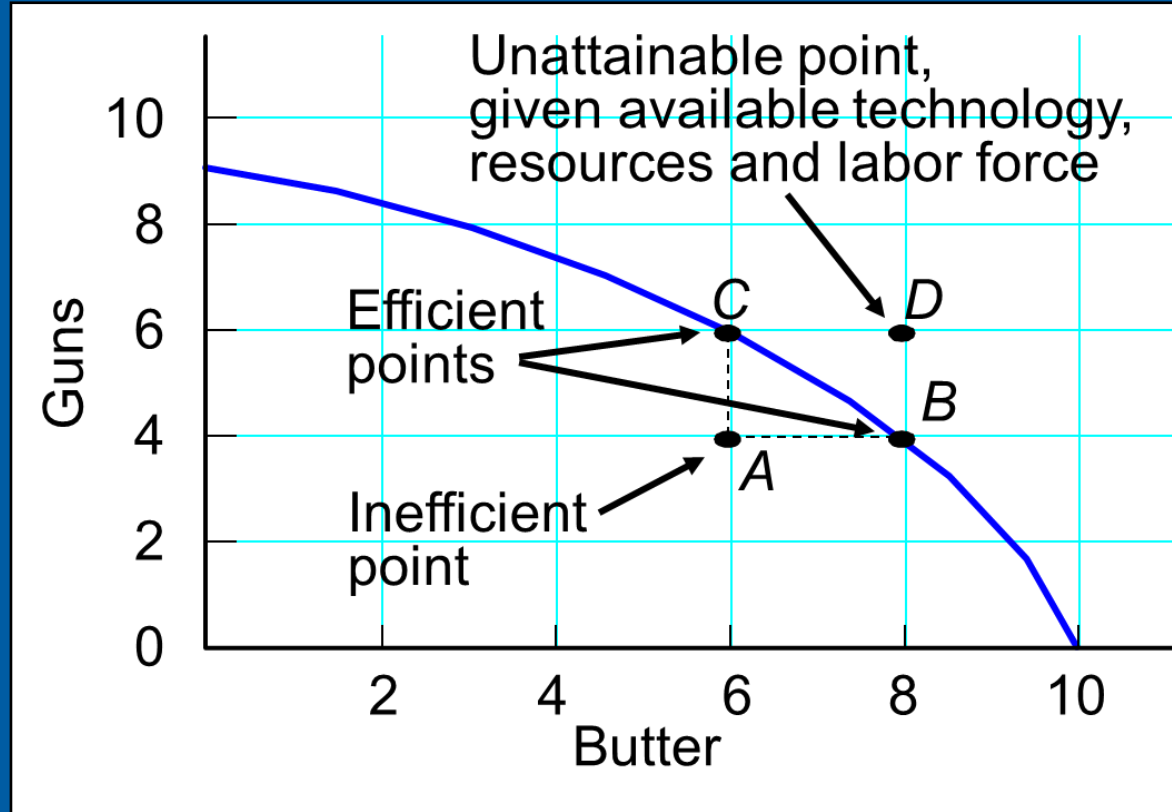
Production Possibilities Model



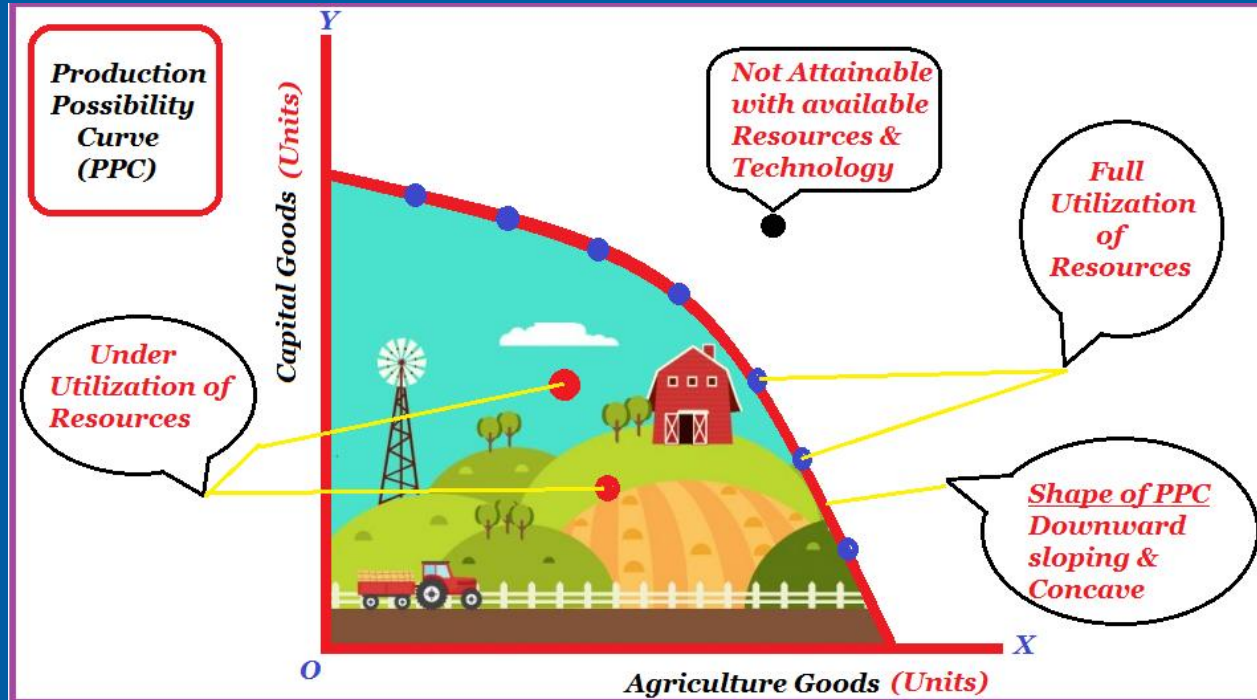
Production Possibilities Model



Efficiency and Inefficiency



Production Possibilities Model (Full Picture)



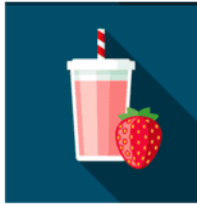
Opportunity Cost

- Opportunity Cost: the loss of all possible alternatives when one good is chosen (decision is made).
 - *Total*: trade-off between the selected good and all the other goods
 - *Marginal*: trade-off between a small part (or unit) of the selected good and a possible alternative
 - *Average*: how much, on average, of one good is sacrificed to produce a possible alternative



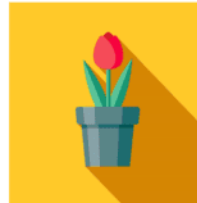
Every decision involves trade-offs

Today's choice



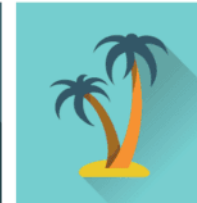
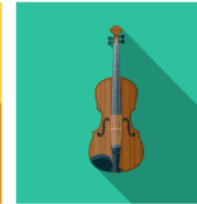
Buying a \$7
strawberry
smoothie

Now



Spending \$7 another way

Later



Benefiting from \$7 saved regularly

Inspired by Andrea Caceres-Santamaria, "Money and Missed Opportunities." *Page One Economics*, October 2019. Icons by Getty Images.

FEDERAL RESERVE BANK *of* ST. LOUIS



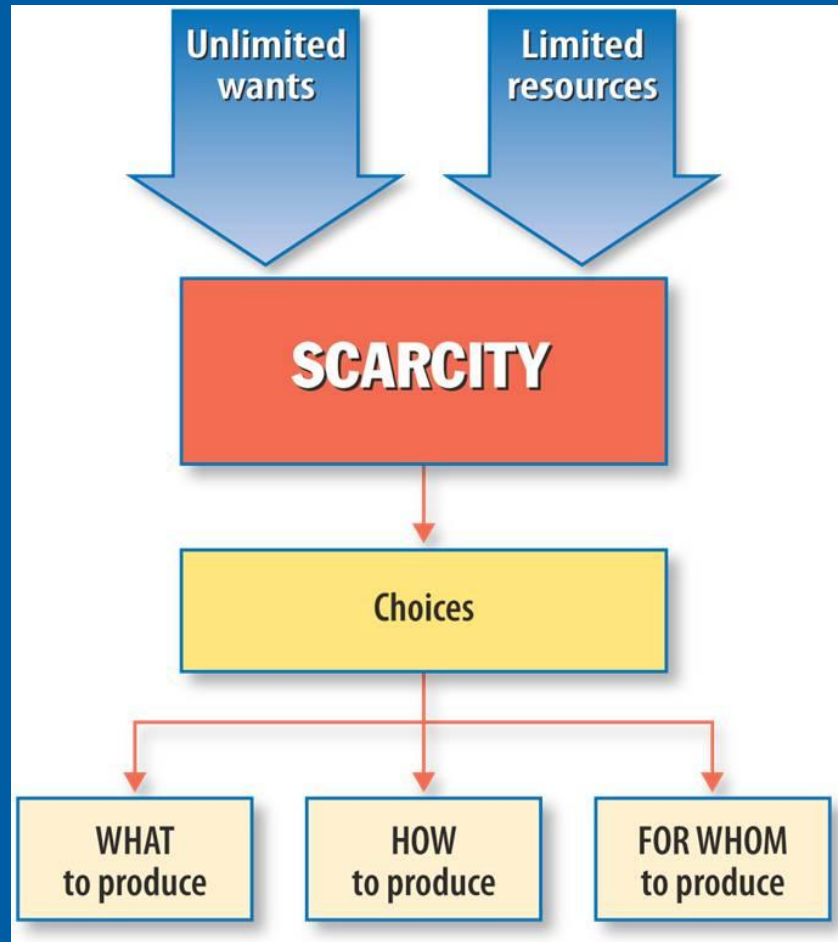
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Opportunity Cost

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Characteristics of the Production Possibility Frontier (PPF)

1. *Downward sloping*

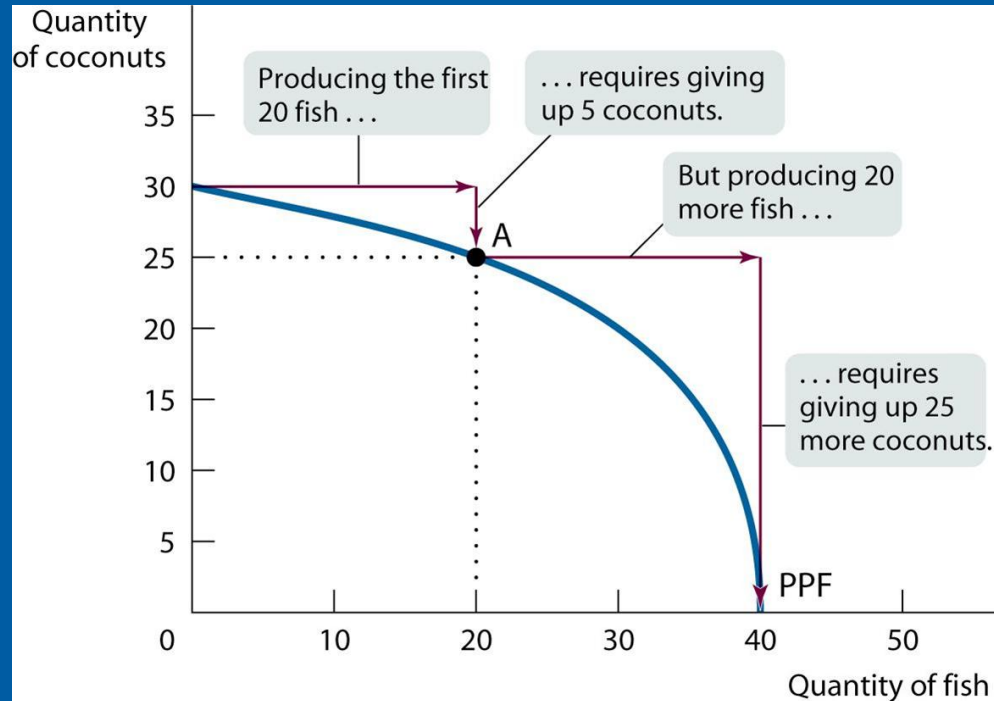
- It is given by the inverse relationship between changes in quantity of two goods
- Remember: PPF shows all the best possible combinations of two goods, which can be produced given the available resources and technology
- Therefore, more of one good can be produced only by taking resources away from the production of another good.

2. *Concave shaped*

- Increasing marginal opportunity costs = more and more units of one good are sacrificed to gain an additional unit of another good.



Increasing Opportunity Cost

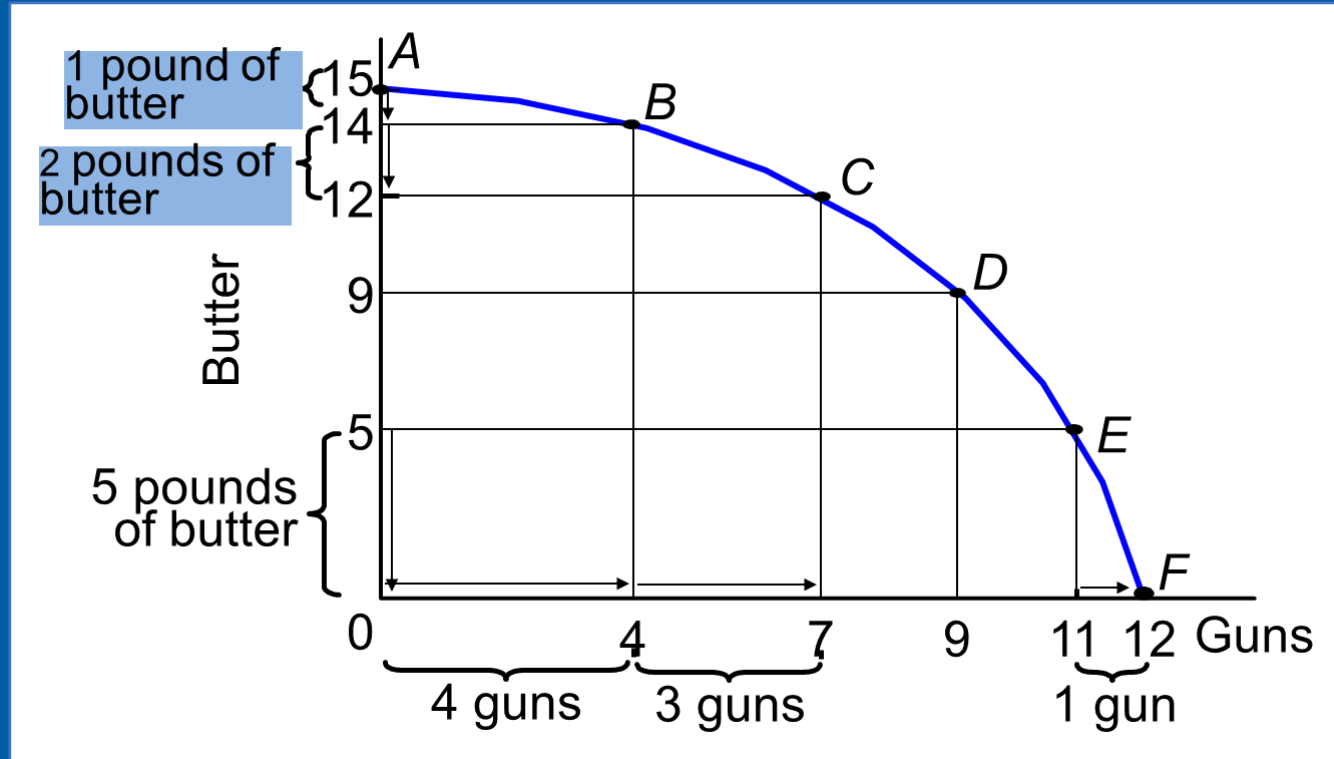


Production Possibilities Table

% of resources devoted to production of guns	Number of guns	% of resources devoted to production of butter	Pounds of butter	Point
0	0	100	15	<i>A</i>
20	4	80	14	<i>B</i>
40	7	60	12	<i>C</i>
60	9	40	9	<i>D</i>
80	11	20	5	<i>E</i>
100	12	0	0	<i>F</i>



Production Possibilities Curve



Example

- 30 days in a month
- two workers: David and Sherry
- two products: Wood and Food
- David can produce 8 unit of food or 4 units of wood per day (or any other linear combination of food and wood).
- Sherry can produce 10 units of food or wood per day (or any other linear combination of food and wood).



Assumptions

- We can produce any positive **real** number of units including (for example: 2.35 units)
- Since products can be measured in any number of ways (pounds, grams, packages, weight, volume etc.) we will simply describe the items in terms of the total number of **units**.



	food	wood
David	8	4
Sherry	10	10

Economists define “**absolute advantage**” as the ability to produce a good or service in greater quantity for the same cost, or the same quantity at a lower cost, than other producers

Who has absolute advantage in producing food? 🍔

Who has absolute advantage in producing wood? 🎄





	food	wood
David	8	4
Sherry	10	10

 $10 > 8$





	food	wood
David	8	4
Sherry	10	10

 $10 > 4$



Similarly, Economists define “**comparative advantage**” as the ability to produce a good or service at a lower opportunity cost than the other producer

Who has comparative advantage in producing food? 🍔

Who has comparative advantage in producing wood? 🎄





	food	wood
David	8	4
Sherry	10	10

David:

Comparative advantage for food  $= \frac{4}{8} = 0.5$

Comparative advantage for wood  $= \frac{8}{4} = 2$



	 food	 wood
David	8	4
Sherry	10	10

Sherry:

Comparative advantage for food  $= \frac{10}{10} = 1$

Comparative advantage for wood  $= \frac{10}{10} = 1$



David 🍔 = 0.5 < 1 = Sherry 🍔

Sherry 🌲 = 1 < 2 = David 🌲

Therefore, Sherry has a comparative advantage in producing wood 🌲 while David has a comparative advantage in producing food 🍔



Daily production		
	Wood	Food
Sherry	10	10
David	4	8

What is the monthly production possibilities frontier?

On aggregate, how much food can David and Sherry produce?

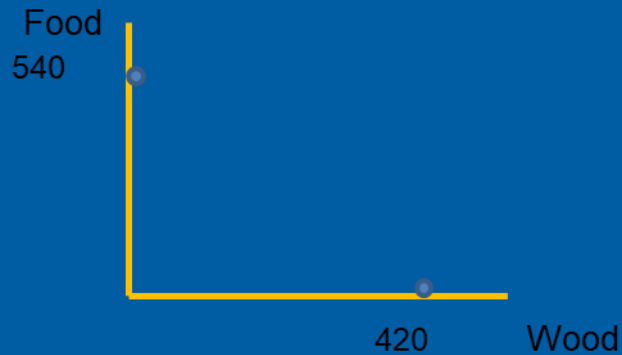
How much wood?



	food	wood
David	8	4
Sherry	10	10

Max Food: $10 \cdot 30 + 8 \cdot 30 = 300 + 240 = 540$

Max Wood: $10 \cdot 30 + 4 \cdot 30 = 300 + 120 = 420$



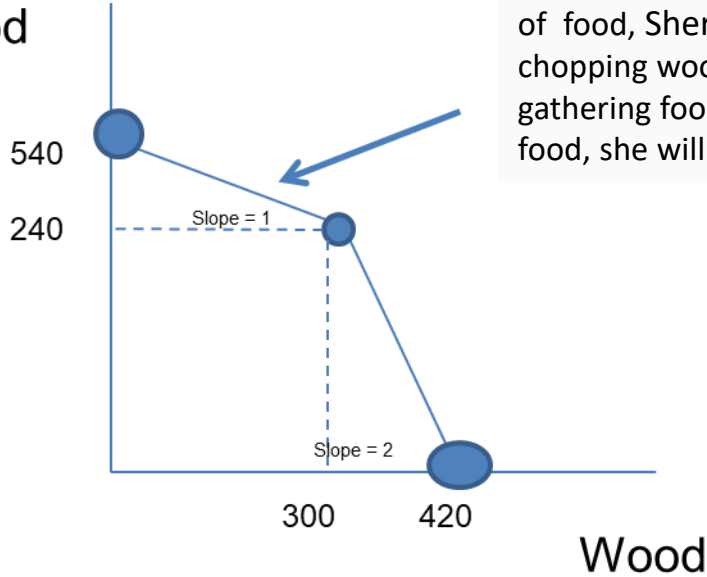
Supposed we are producing just 420 pieces of wood, but we need some food. Who is going to produce the food? David or Sherry ?

⇒ Key answer: Opportunity cost

	Wood	Food	Opportunity cost of Wood	Opportunity cost of Food
Sherry	10	10	1	1
David	4	8	2	0.5



Food



If they need more than 240 units of food, Sherry should stop chopping wood and start gathering food. For 1 unit of food, she will lose 1 unit of wood.

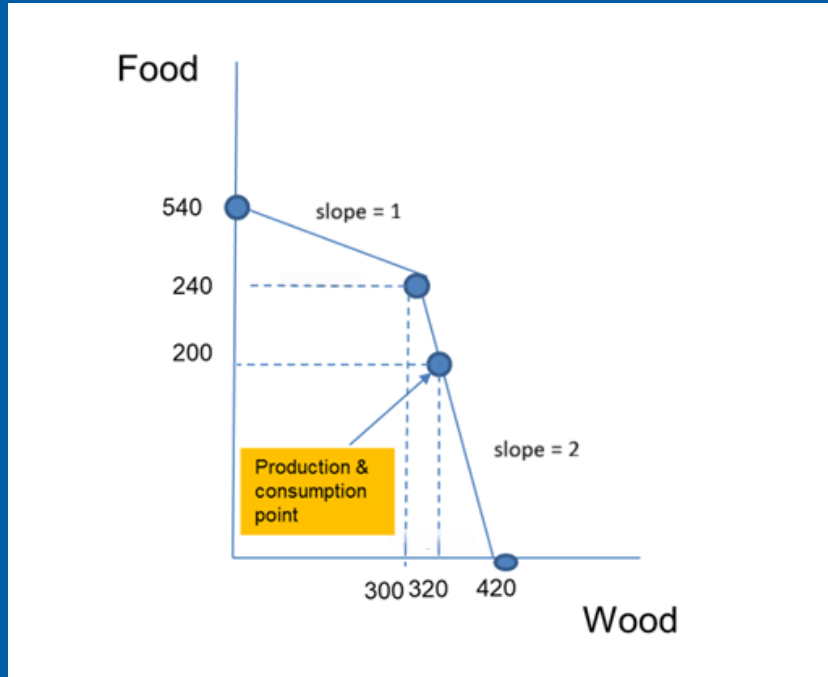
David can gather 240 units of Food instead of 120 units of Wood.

In this case, they can produce 240 (30×8) units of food and 300 (30×10) of wood.

Note: David can decide how many days to spend cutting wood and how many days to gather food. The cost opportunity is always 0.5 wood (Linear line)



Suppose they need 200 units of food and as many of woods as they can get. How many units of wood can they produce ?



David has a comparative advantage in producing food. He can spend 25 days $(200/8)$ producing food and 5 days chopping woods. Sherry is going to just chop woods. Therefore, they produce 200 units of food and 320 $((4*5)+(10*30))$ units of wood.



TRADE - Assumptions

- No transaction costs
- Same products are exchanged in the market
- Producers cannot affect the market price (no market power)
- Producers can sell and buy unlimited quantities from/to the world market.



TRADE #1

- Suppose David and Sherry can trade 1 unit of food for 2 units wood and that they still need 200 units of food and as many woods as they can get. Find the production and consumption point.

	Wood	Food	Opportunity cost of producing Wood	Opportunity cost of producing Food	Opportunity cost of trading Wood	Opportunity cost of trading Food
Sherry	10	10	1	1	$\frac{1}{2} = 0.5$	$2/1 = 2$
David	4	8	2	0.5	$\frac{1}{2} = 0.5$	$2/1 = 2$



Food

Production
point

540

slope = 1

240

200

slope = 2

Consumption
and production
point without
trade

300

320

420

680

Wood

Consumption point

The opportunity cost of producing 1 food is either 1 or 0.5. In both cases the opportunity cost is lower than in trading. Therefore, David and Sherry will produce only food (540 units of food) and sell the extra 340 units of food for 680 units of wood.



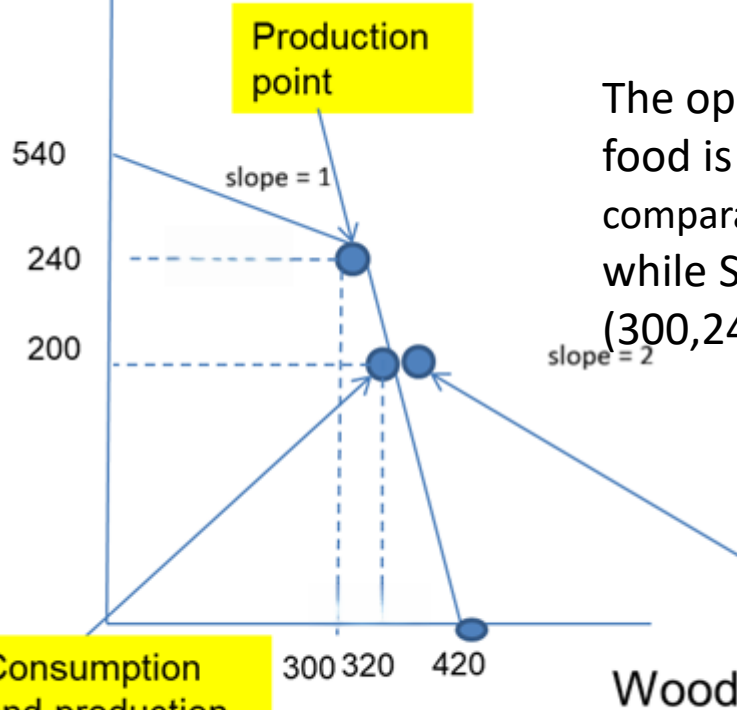
TRADE #2

Suppose they now trade 1 unit of food for 0.75 of wood

	Wood	Food	Opportunity cost of producing Wood	Opportunity cost of producing Food	Opportunity cost of trading Wood	Opportunity cost of trading Food
Sherry	10	10	1	1	$1/0.75 = 1.33$	$0.75/1 = 0.75$
David	4	8	2	0.5	$1/0.75 = 1.33$	$0.75/1 = 0.75$



Food



The opportunity cost of producing 1 unit of food is either 1 or 0.5. David has a comparative advantage in producing food while Sherry in wood. Production point: (300,240). Consumption point: (330,200)

$$330 = 300 + (240 - 200) * 0.75$$

Consumption point



TRADE #3 (Homework)

- DIY (at home) Suppose they can trade 1 unit of food for 0.2 of woods. Find the consumption and production point.

	Wood	Food	Opportunity cost of producing Wood	Opportunity cost of producing Food	Opportunity cost of trading Wood	Opportunity cost of trading Food
Sherry	10	10	1	1	?	?
David	4	8	2	0.5	?	?



Summary #1

1. The field of economics is divided into two subfields: microeconomics and macroeconomics.
2. Economists try to address their subject with scientific objectivity.
3. A positive statement is an assertion about how the world is.
4. A normative statement is an assertion about how the world should be.
5. The production Possibilities Frontier is a graph that shows all the combinations of goods and services that can be produced if all the society's resources are fully employed and used efficiently.



Summary #2

1. The cost of something is what you give up to get it. This is called the opportunity cost.
2. A producer has a comparative advantage over another in the production of a good or service if he/she can produce the good or service at lower opportunity cost.
3. A producer has an absolute advantage over another in the production of good or service if he/she can produce the good or service using fewer resources.

